



SESSION 2025/2026, SEMESTER 1

SECJ3623: MOBILE APPLICATION PROGRAMMING

SECTION: 01

FINAL PROJECT REPORT

“EASYINVO – SIMPLE INVOICE MANAGEMENT APPLICATION”

PREPARED FOR:

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1.0 INTRODUCTION

Invoicing is a critical operational activity for small businesses and freelancers, as it directly affects cash flow, financial tracking, and professional communication with clients. However, many small enterprises continue to rely on manual or semi-digital methods such as handwritten invoices, spreadsheets, or word processing documents. These approaches are often time-consuming, prone to human error, and inefficient when it comes to tracking payments, managing client records, or generating financial summaries. As businesses grow, these limitations can lead to poor financial visibility and increased administrative burden.

EasyInvo is a proposed invoice management system designed to address these challenges by providing a centralized, user-friendly digital solution for creating, managing, and tracking invoices. The system is primarily targeted at small business owners, freelancers, and micro-enterprises who require a simple yet effective invoicing tool without the complexity of large-scale accounting software. EasyInvo focuses on ease of use, automation, and clarity, ensuring that users with minimal technical or accounting background can manage their invoicing tasks efficiently.

The EasyInvo system enables users to create professional invoices, store and reuse client and item information, generate invoice summaries, and monitor payment status in real time. By digitizing the invoicing workflow, the system reduces repetitive manual work and minimizes errors related to calculations, missing information, or inconsistent formatting. Additionally, EasyInvo supports invoice sharing through commonly used communication channels and provides downloadable invoice documents in a professional format, enhancing credibility and trust between businesses and their clients.

This report presents a comprehensive overview of the system design and proposed solution for the EasyInvo project. It outlines the identification of stakeholders and users, summarizes insights gathered from stakeholder meetings, and describes the system's functional scope using UML-based modeling. The report also discusses the overall system flow, navigation structure, and interface design approaches, including both low-fidelity and high-fidelity designs. Finally, it explains how the proposed design decisions align with real user needs and business requirements, ensuring that EasyInvo delivers practical value as a lightweight and effective invoicing solution.

2.0 STAKEHOLDERS AND USER IDENTIFICATION

This section provides a detailed overview of the key stakeholders and users of the **EasyInvo** system. Understanding the needs and expectations of these parties is critical for ensuring that the system meets real-world requirements, supports business workflows, and delivers a user-friendly invoicing experience. Stakeholders are individuals or groups who influence, use, or are impacted by the system, while users are the direct operators who interact with EasyInvo in their daily tasks.

2.1 Stakeholders

Stakeholders play an important role in shaping the functional and non-functional requirements of EasyInvo. Their feedback guides the design of core features, ensures the system aligns with business needs, and helps validate usability and efficiency.

Stakeholder Assignment Overview

Stakeholder Type	Assigned Team Member(s)	Task / Action
Small Business Owner	Janavi, Aimie	Conduct interviews to understand operational challenges and invoicing needs
Clients / Customers	Elawarasi, Faiz	Gather feedback on invoice clarity, usability, and overall experience

2.2 User Identification

EasyInvo is designed for multiple user roles, each with specific responsibilities and access levels.

Business Owner / Freelancer (Primary User)

- Manages the business profile, client database, and invoice records
- Creates and sends invoices, tracks payments, and monitors financial summaries
- Ensures timely follow-ups and financial accuracy

Administrative Staff (Secondary User)

- Assists with invoice creation and client management
- Updates invoice statuses and maintains accurate customer records
- Supports business operations without full system control

Accountant (Read-Only / Limited Access User)

- Reviews invoices and financial records
- Generates reports for accounting purposes
- Monitors overall financial accuracy while maintaining restricted access to sensitive operations

2.3 Stakeholder Meeting Summaries

Stakeholder 1 – Thirtana (Small Business Owner)

Meeting Date/Time: 27/11/2025, 3:00 PM

Assigned To: Elawarasi

A detailed session was conducted with Thirtana to understand the challenges small business owners face in managing invoices and client information.

Key Discussion Points:

- Manual invoice creation is time-consuming and prone to errors\
- Fast and efficient invoice generation is needed
- Reusable customer profiles would reduce repetitive work
- Business profile setup should allow branding elements (logo, address, contact details)
- Invoice status tracking (paid/unpaid) is essential for financial management

Key Outcomes:

- Core functional requirements were established
- Emphasis on simplicity, automation, and time-saving features
- Insights validated EasyInvo's design approach for small business workflows

Stakeholder 2 – Puan Sharifah (Quzza Enterprise Sdn Bhd)

Meeting Date/Time: 18/11/2025, 12:00 PM

Assigned To: Aimie

The meeting explored invoicing practices within a small construction business, focusing on the limitations of current methods.

Key Discussion Points:

- Invoices are created using Word and Excel, making tracking and organization difficult
- On-site invoice creation is necessary for mobile operations
- Centralized client information storage improves accuracy and efficiency
- Professional PDF invoices with company branding enhance business image
- Simple monthly summaries facilitate payment tracking and financial reporting

Key Outcomes:

- Requirements for mobile-friendly and professional invoice formats were confirmed.
 - The importance of centralized data management was highlighted
-

Stakeholder 3 – Taquiuddin (Customer)

Meeting Date/Time: 19/11/2025, 1:00 PM

Assigned To: Janavi

This session represented the perspective of invoice recipients and end clients.

Key Discussion Points:

- Clear and readable invoice presentation
- Easy access to invoices, including PDF download
- Visibility of key information such as amounts, due dates, and payment status

Key Outcomes:

- Reinforced the need for a clean, simple, and intuitive invoice layout
 - Helped define usability requirements for client-side interaction
 - Feedback confirmed EasyInvo's goal of delivering a professional and user-friendly experience
-

Stakeholder 4 – Mr Firdaus (Customer)

Meeting Date/Time: 15/11, 10am

Assigned To: Faiz

Key discussion points:

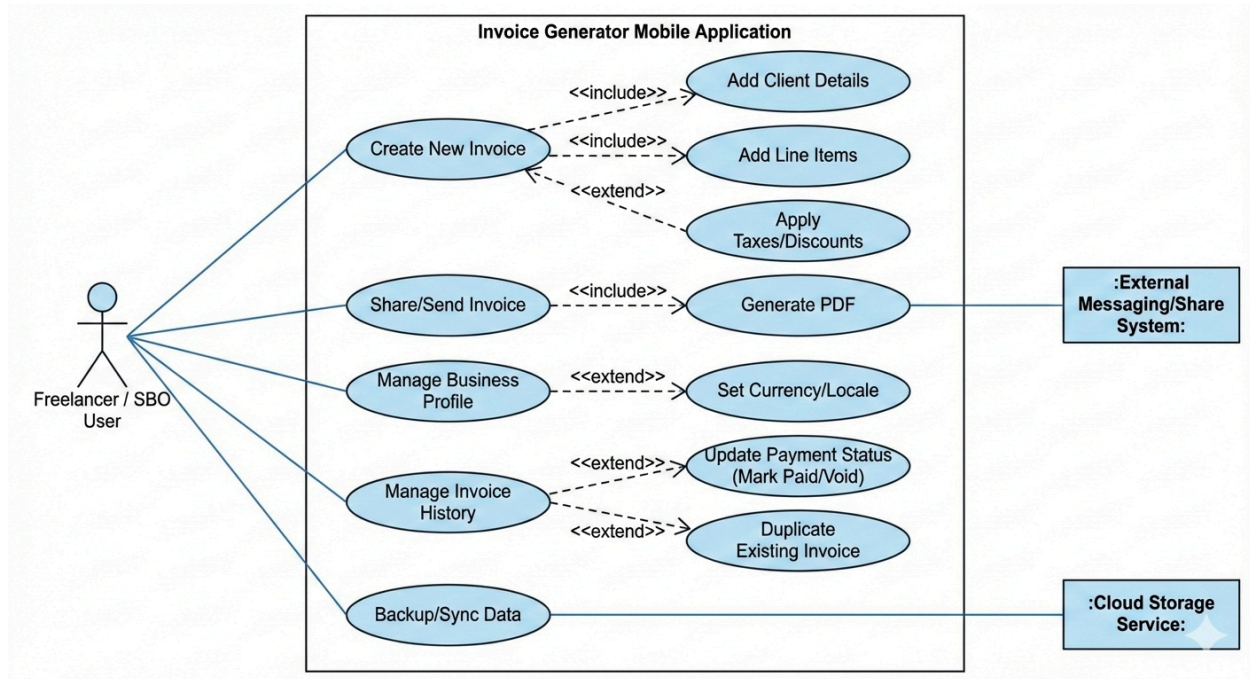
- Streamlined Data Entry & Auto-fill Capabilities
- Customization and Branding Options
- Multi-Channel Sharing Integration

Key outcomes:

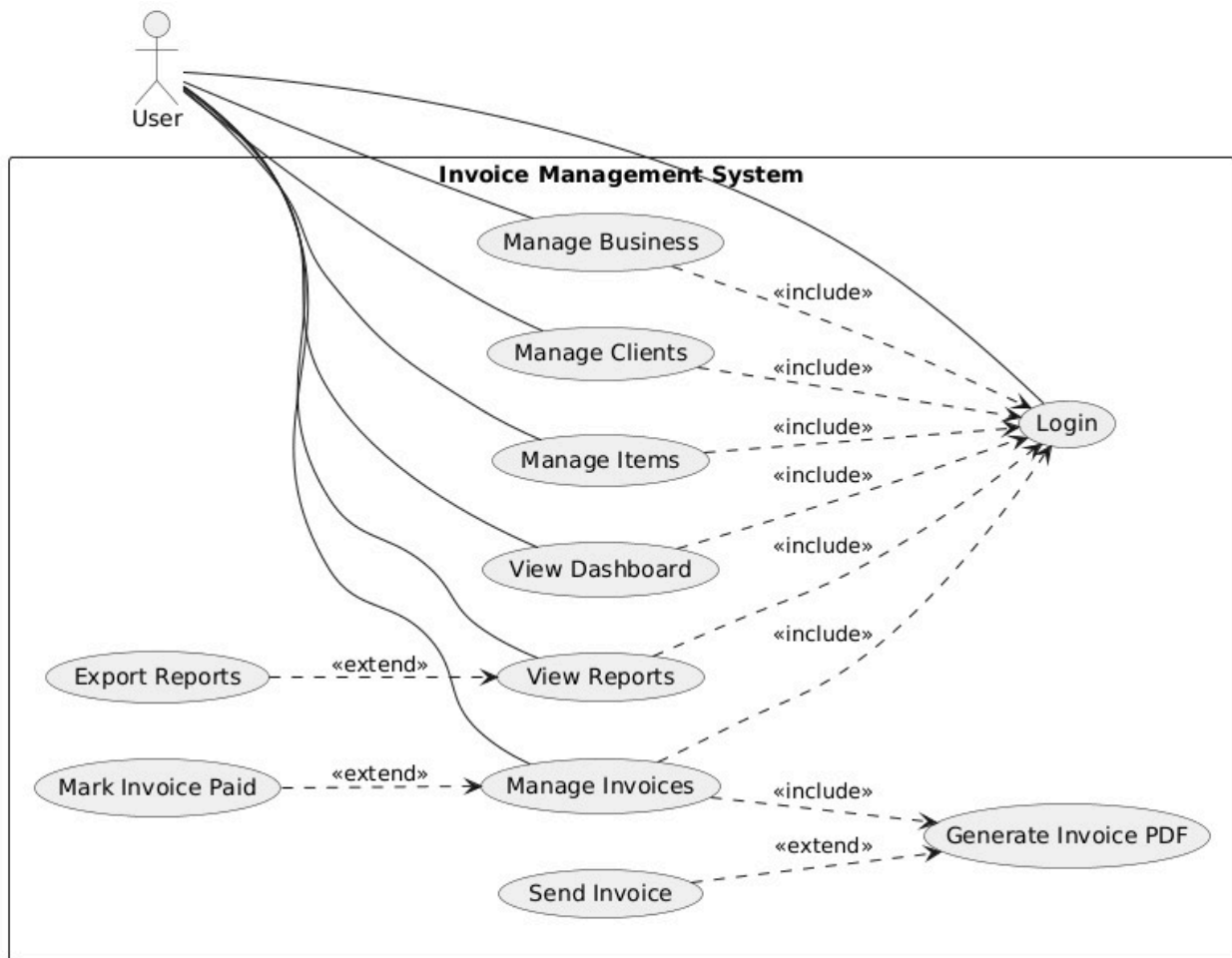
- Established a "Low-Friction" User Flow
- Decided that the app must function without an internet connection
- Confirmed that the app will generate a lightweight PDF link or file that is compatible with instant messaging apps, not just traditional email attachments.

3.0 UML DIAGRAMS

3.1 Actor Diagram

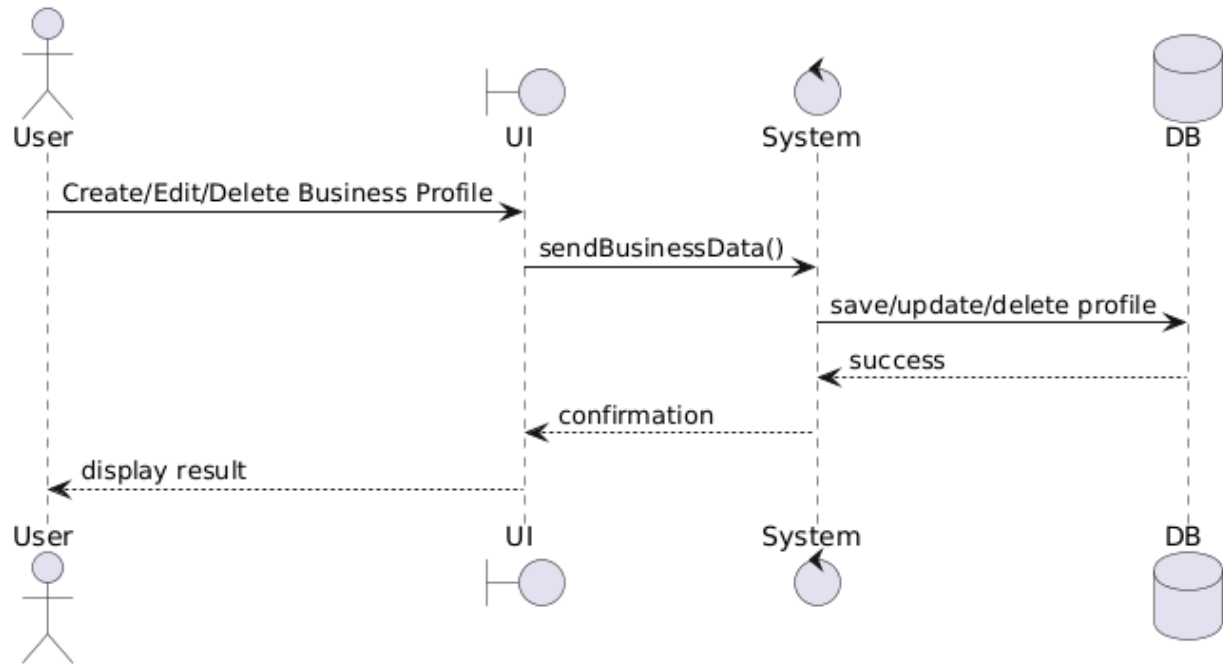


3.2 Use-Case Diagram

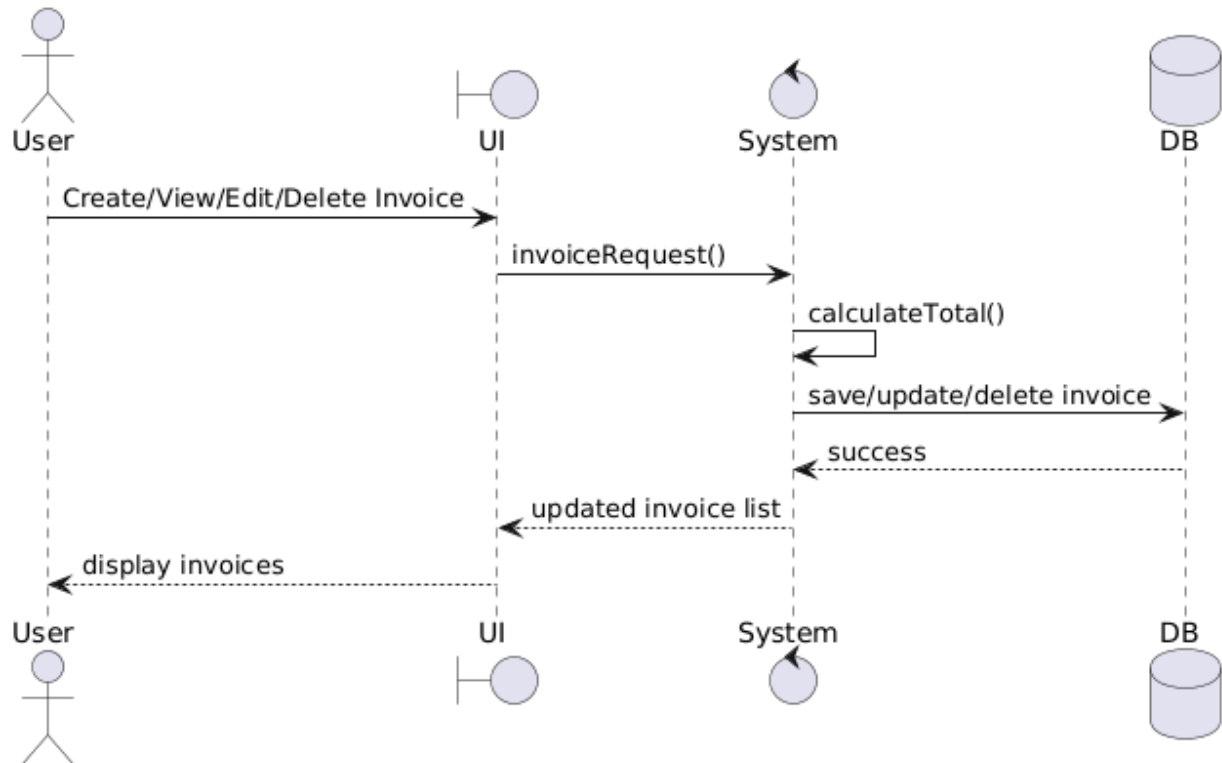


3.3 Sequence Diagram

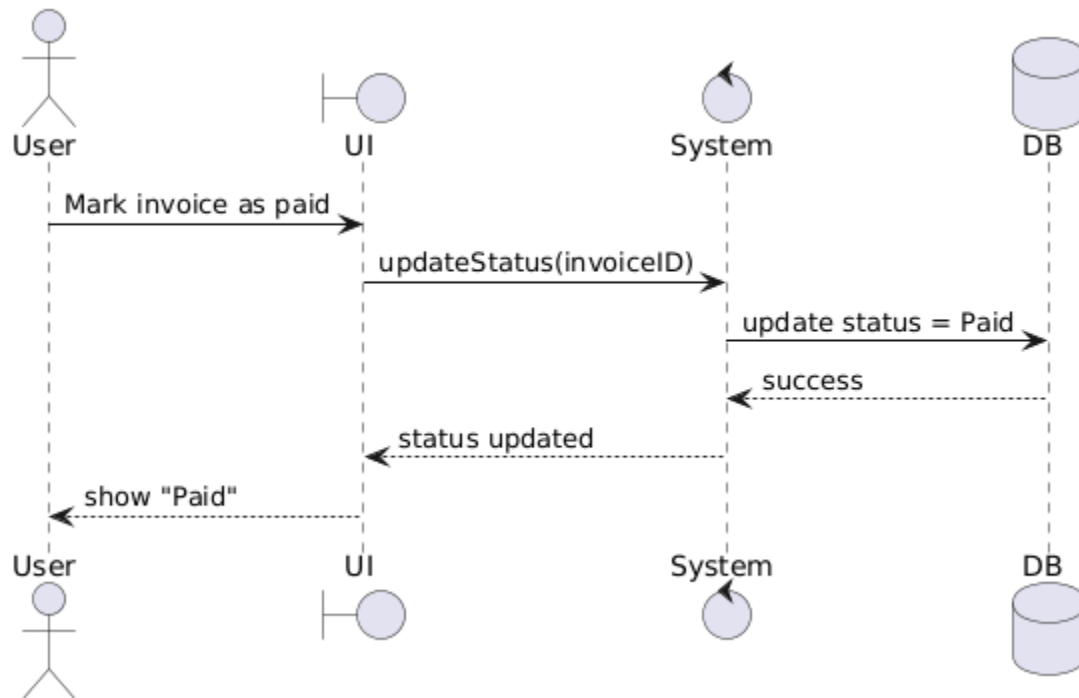
Business Profile Management



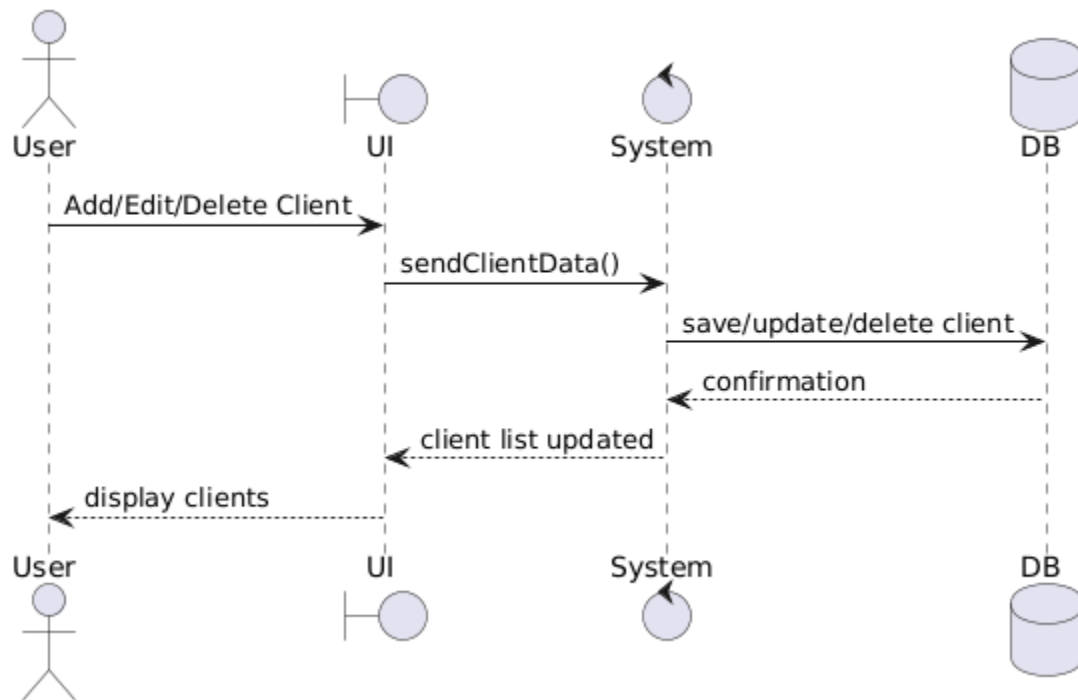
Invoice Management



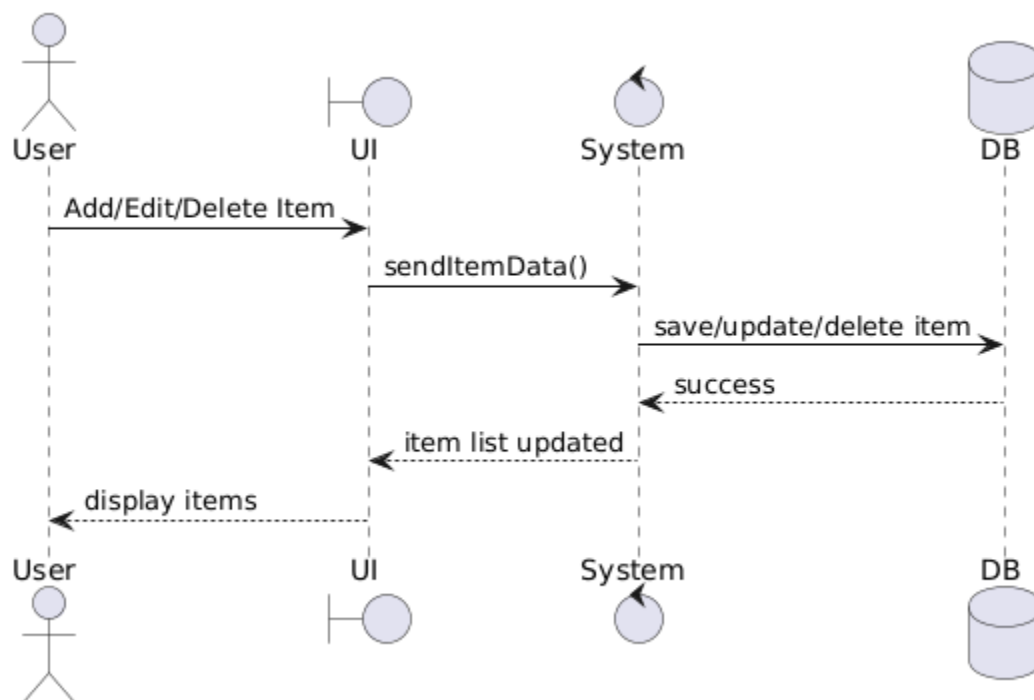
Invoice Paid



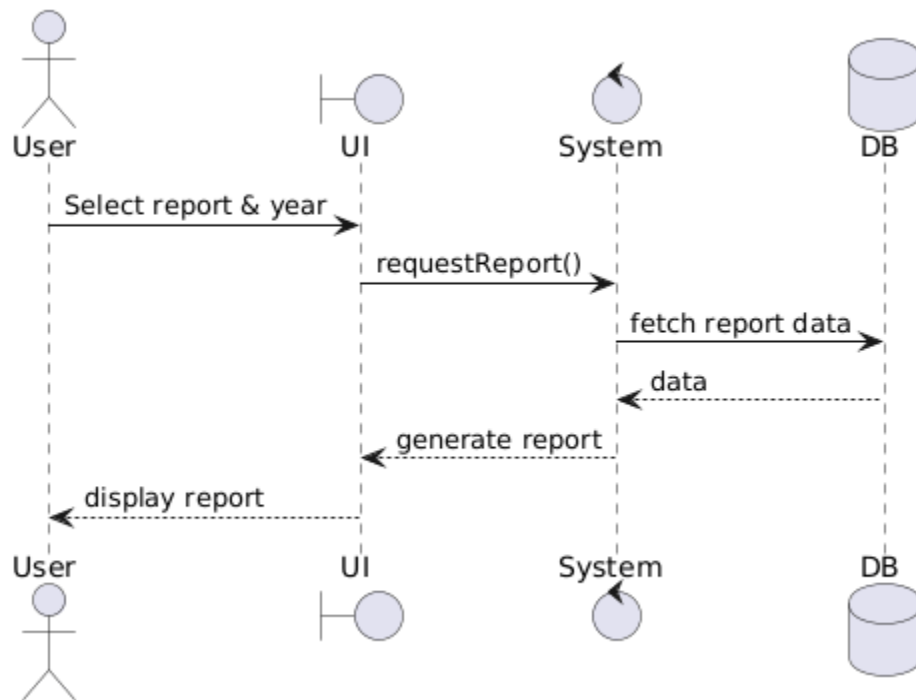
Client Management



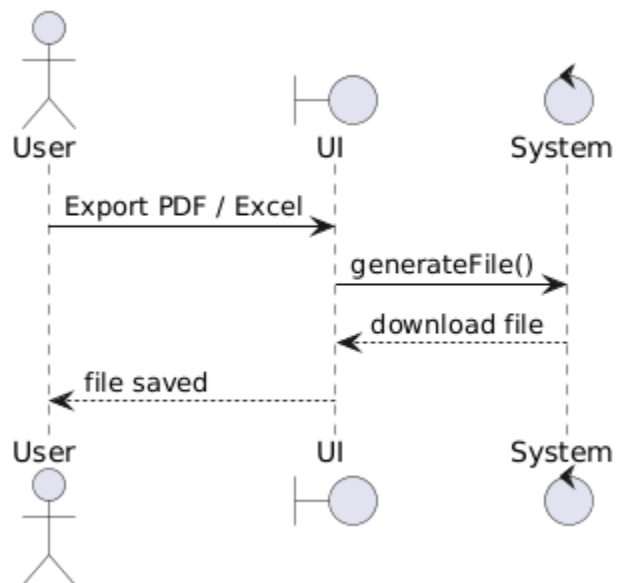
Item / Product Management



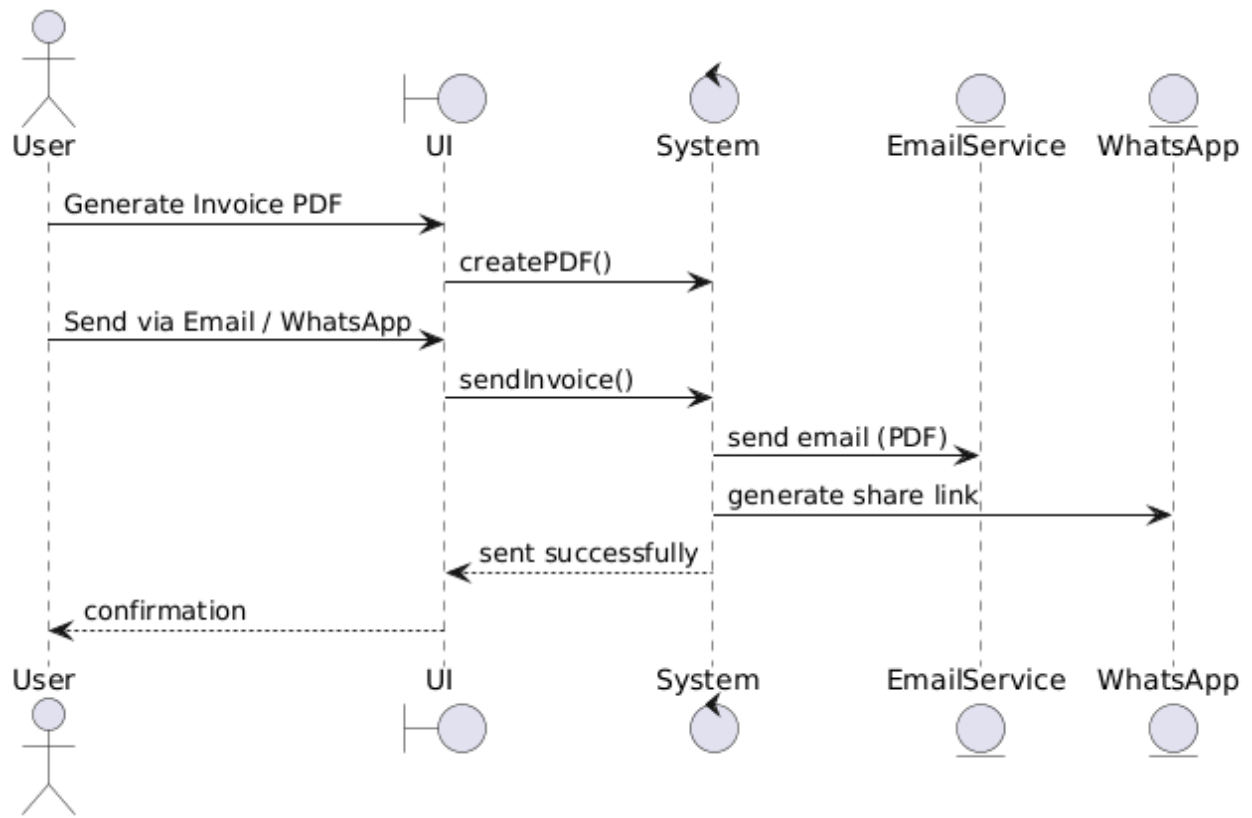
Reports and Analytics



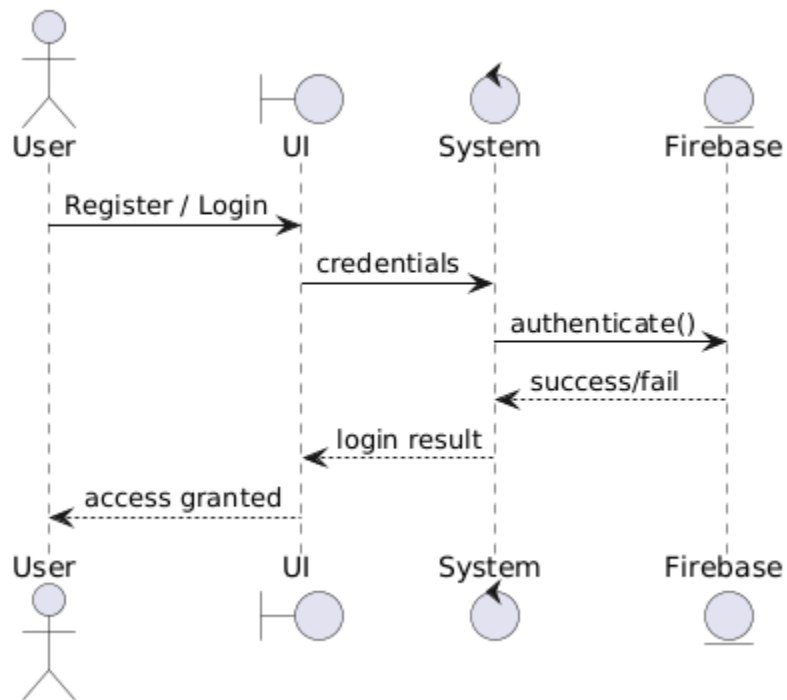
Report Exporting



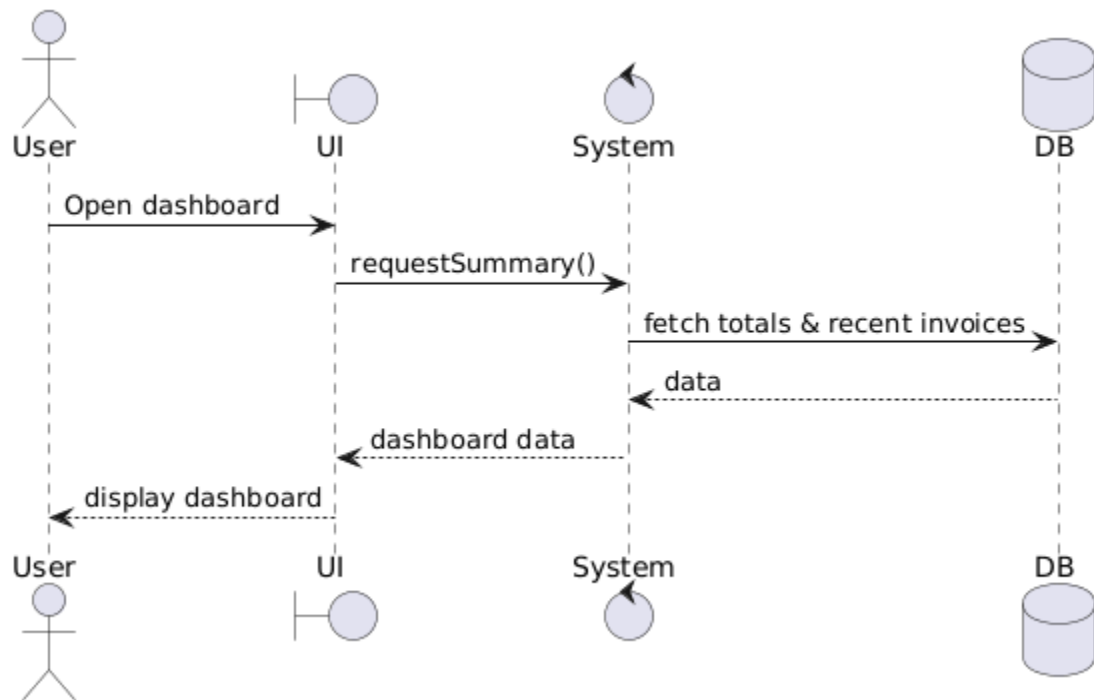
Invoice Generation & Sharing



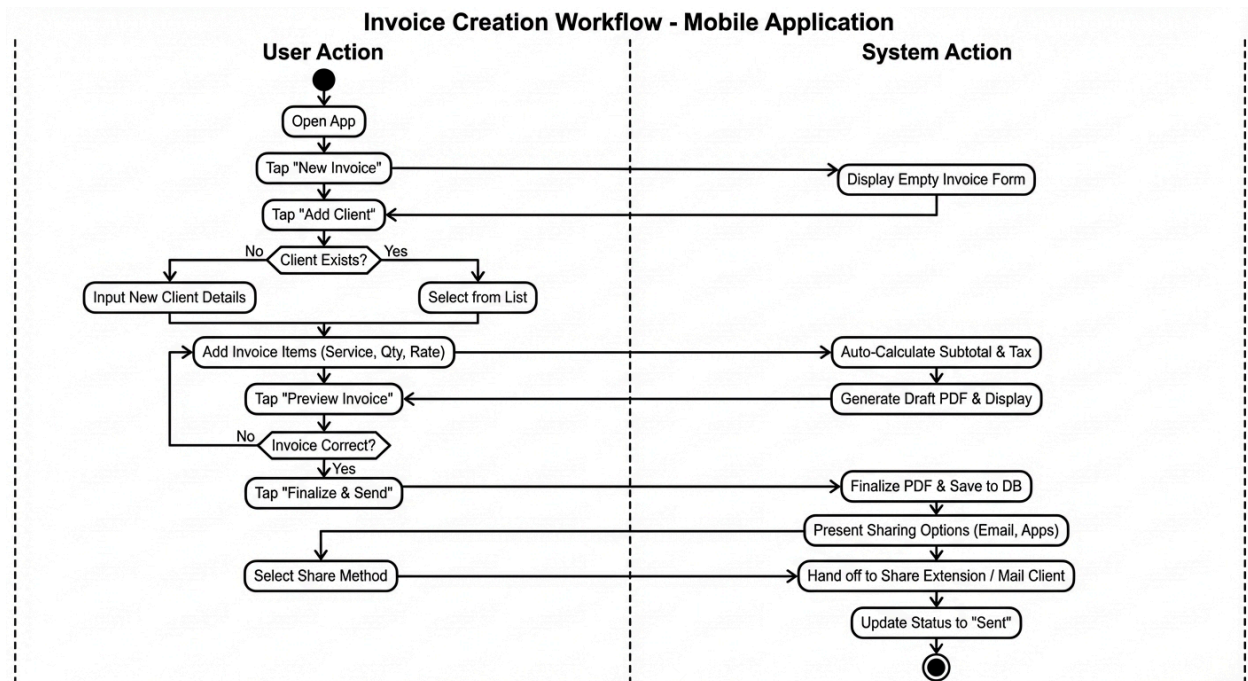
User Authentication



Dashboard Overview



3.4 Activity Diagram



4.0 SYSTEM FLOW AND NAVIGATION STRUCTURE

4.1 System Flow

The system flow of EasyInvo illustrates how users interact with the application from initial access to completing various invoicing and management tasks. The flow is designed to be intuitive, efficient, and user-friendly, minimizing unnecessary steps while ensuring accurate data entry and management.

User Authentication

The system begins when a user accesses EasyInvo and is presented with the **landing screen**. Here, users can:

- Log in using existing credentials
- Register for a new account
- Reset their password if they have forgotten it

Successful authentication grants access to the system, directing users to the **dashboard** based on their role and access level.

Dashboard Access

After logging in, the user is directed to the **dashboard**, which serves as the main navigation hub. The dashboard provides:

- A high-level overview of key information, such as total invoices, unpaid invoices, and recent activity
- Quick access to all modules, including Invoice Management, Business Profile, Customer Management, Products, Income Overview, Reports, and Settings

The dashboard allows users to efficiently navigate to any app feature with minimal effort.

Business Profile Management

For first-time users, the system prompts the creation of a **business profile**. This includes:

- Company name, address, and contact details
- Company logo for branding purposes

These details are automatically applied to all invoices, ensuring consistency and professional presentation. Users can also edit or delete existing business profiles and add new profiles as needed.

Customer Management

Users can manage customer information within the **Customer Management** module. This includes:

- Viewing a list of all customers
- Adding new customer records
- Editing or deleting existing customer details

Centralized customer management allows for quick selection during invoice creation, reducing repetitive data entry.

Product / Item Management

The **Product Page** allows users to manage products or services offered by their business. Users can:

- View all products added
- Add new products with descriptions, pricing, and stock information
- Edit or delete existing products
- Search for specific products

Storing product information in the system streamlines invoice creation and ensures accurate calculations.

Invoice Creation and Management

Users create invoices by selecting a customer, adding products or services, and specifying quantities and prices. The system automatically calculates totals, taxes, and final amounts to minimize errors.

Once created, users can:

- Review invoices before finalizing
- Mark invoices as **paid** or **unpaid**
- Filter invoices by status (paid/unpaid)
- Delete or edit invoices as needed
- Share invoices via PDF or other supported channels
- Print invoices directly from the app

This module ensures end-to-end management of all invoice-related tasks in one place.

Income Overview

The **Income Page** provides users with a summary of their financial performance. Users can view:

- Total income
- Breakdown by payment methods, such as cash, card, bank transfer, mobile payments, and others

This allows businesses to quickly assess revenue sources and monitor payment trends.

Reports and Analytics

Users can generate detailed reports to analyze business performance. Reports include:

- All invoices or filtered by paid/unpaid status
- Customer-specific activity
- Product-specific details
- Annual reports filtered by year

Reports can be exported in **PDF or Excel** formats, supporting financial tracking, accounting, and decision-making.

Settings and Customization

The **Settings** module allows users to:

- Customize invoice templates for professional branding
- Change account passwords
- Access help and support
- View application information

This ensures that users can personalize the app to match their business requirements and maintain account security.

Logout

After completing tasks, users can securely log out, ensuring the safety of business and customer data.

EasyInvo System Flow

User Journey from Start to Finish



4.2 Navigation Structure

The navigation structure of EasyInvo is organized to ensure ease of access and logical grouping of features. The EasyInvo mobile application is designed with a user-friendly navigation structure to allow seamless access to all features.

1. Landing Screen (Login / Registration)

- The entry point of the app where users can either log in with existing credentials or create a new account.
- A “Forgot Password” option is available for users to reset their password if needed.

2. Dashboard

- Serves as the home screen of the app.
- Provides navigation to all major functions of the app via buttons or menu options.
- Users can access invoice management, business profiles, customer management, product page, income overview, reports, and settings from this central location.

3. Invoice Management

- The main invoice page provides an overview of all invoices, showing their status (paid or unpaid).
- Users can:
 - Filter invoices by paid or unpaid status
 - Create, edit, view, or delete invoices
 - Mark invoices as paid or unpaid
 - Share invoices via PDF or other sharing options
 - Print invoices directly from the app

4. Business Profile

- Displays an overview of all business profiles added by the user.
- Users can:
 - Add new business profiles
 - Edit existing business profiles
 - Delete business profiles

5. Customer Management

- Provides an overview of all customers associated with the business.
- Users can:
 - Add, view, edit, or delete customer details

6. Product Page

- Shows all products added by the user.
- Users can:
 - Add new products and product descriptions
 - Update stock and pricing information
 - Delete products
 - Search for products
 - View product details

7. Income Overview

- Provides reports on total income and breakdowns by payment method (e.g., cash, card, credit card, mobile payments, bank transfer, check, debit, others).

8. Reports

- Users can generate detailed reports including:
 - All invoices or filtered by paid/unpaid status
 - Customer-specific reports
 - Product-specific reports
 - Annual reports filtered by year
- Reports can be exported in PDF or Excel format.

9. Settings

- Allows users to customize invoice templates
- Change account password
- Access help and support resources
- View app information

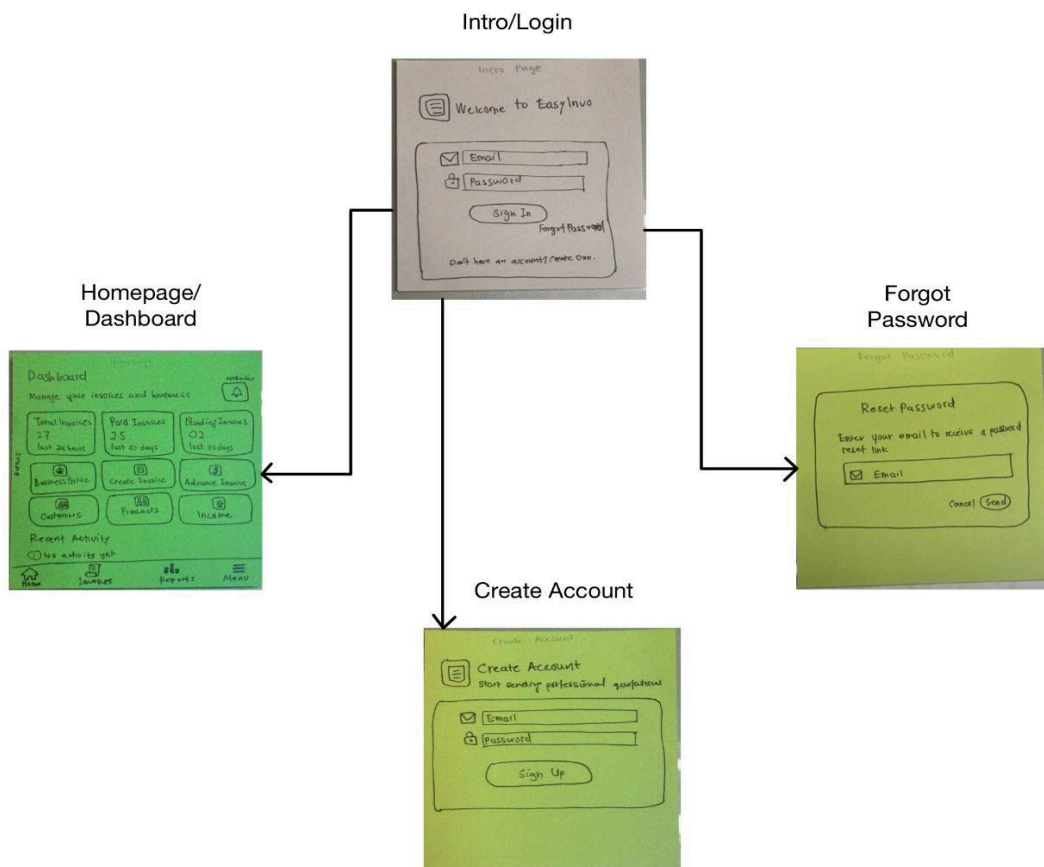
This structured navigation ensures users can easily locate features and complete tasks with minimal effort.

5.0 INTERFACE DESIGN

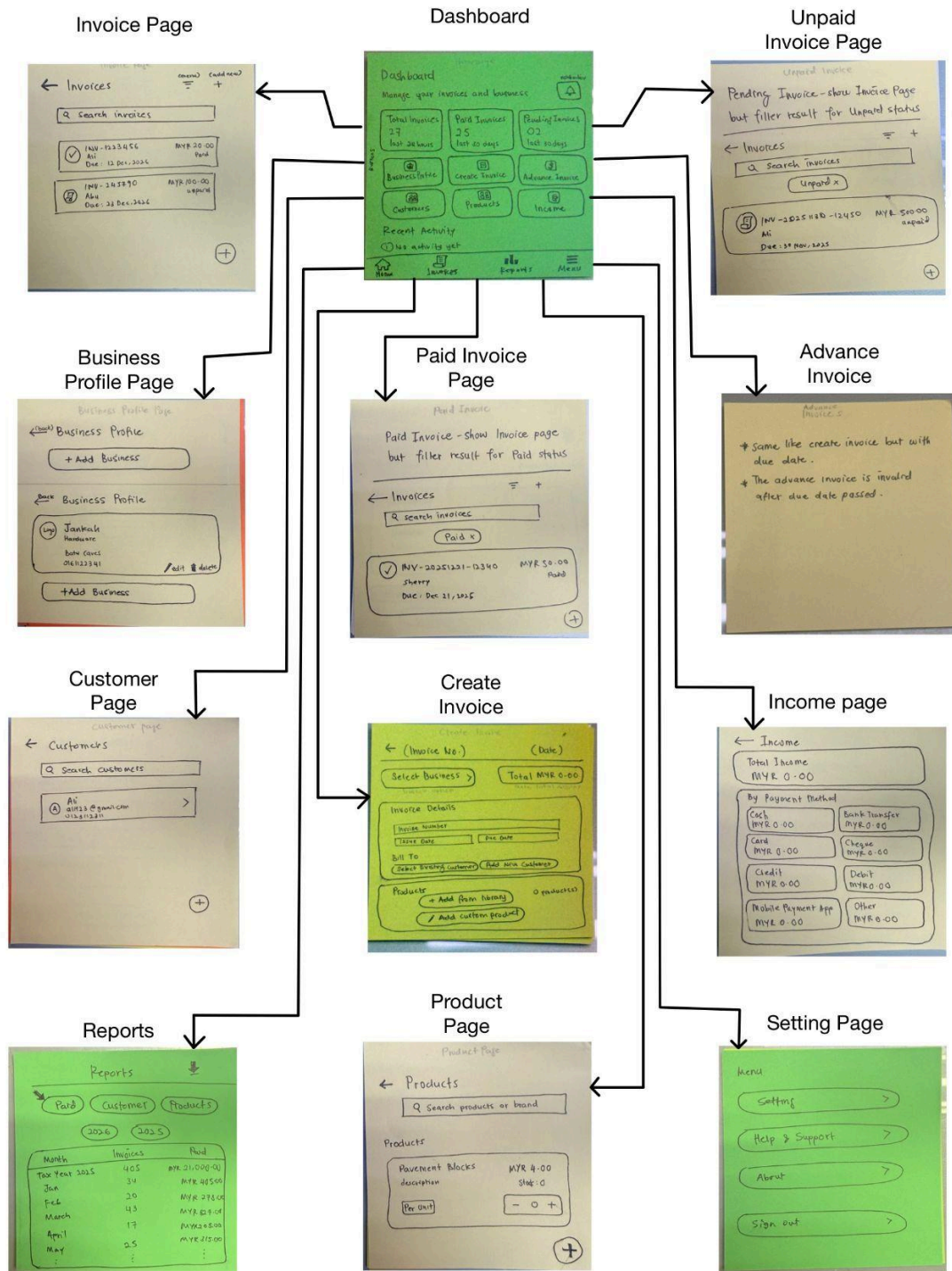
5.1 Low Fidelity

EASYINVO LOW FIDELITY WIREFRAME

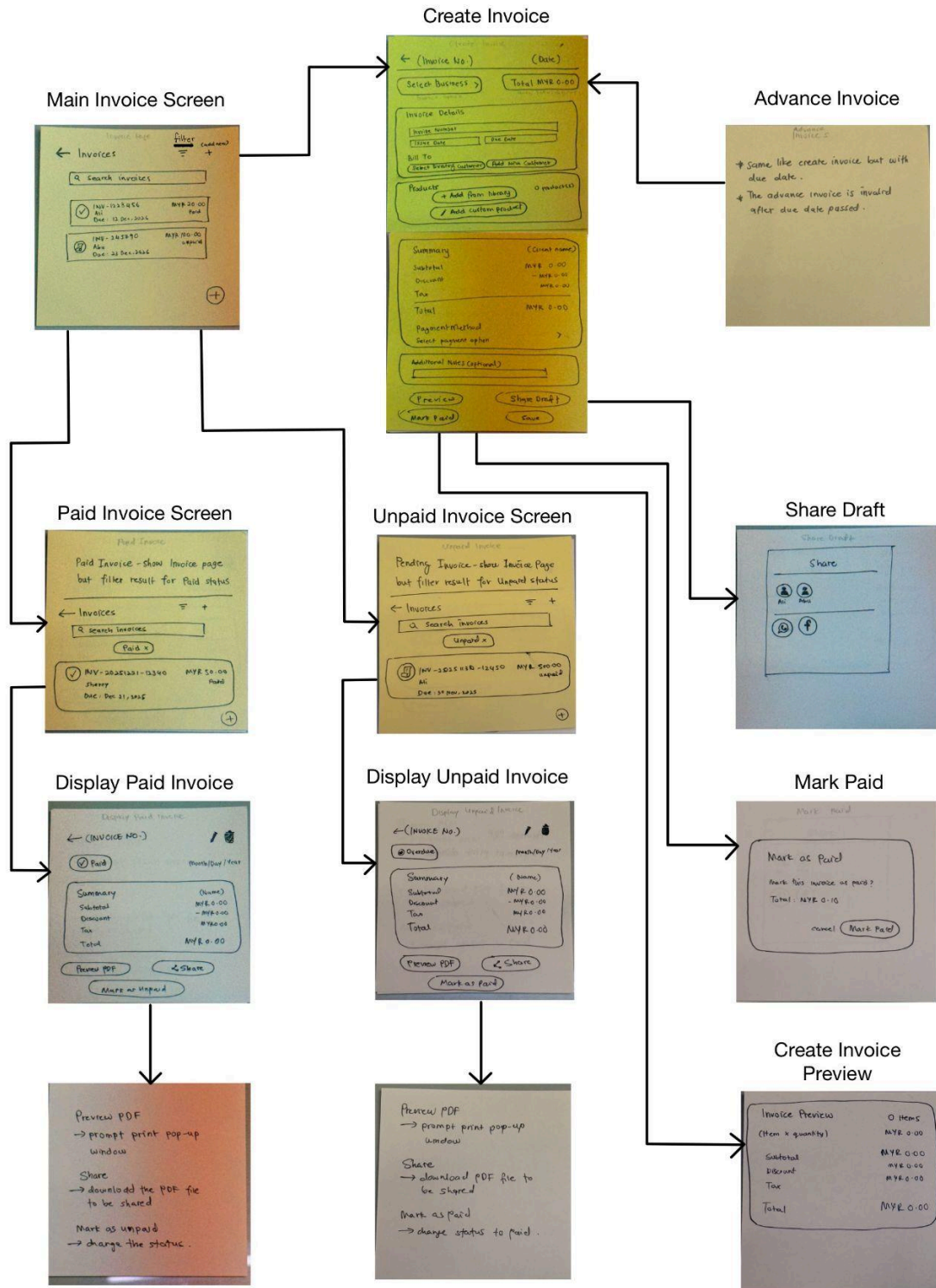
General Screen



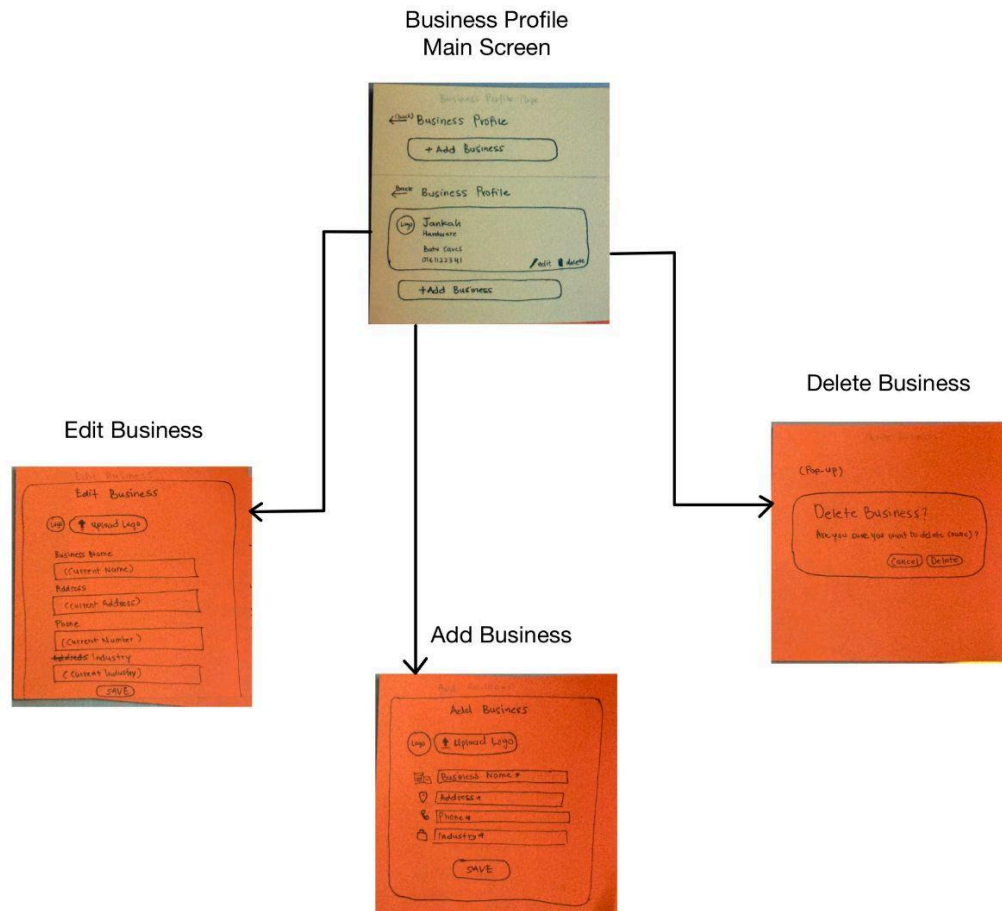
Home Screen



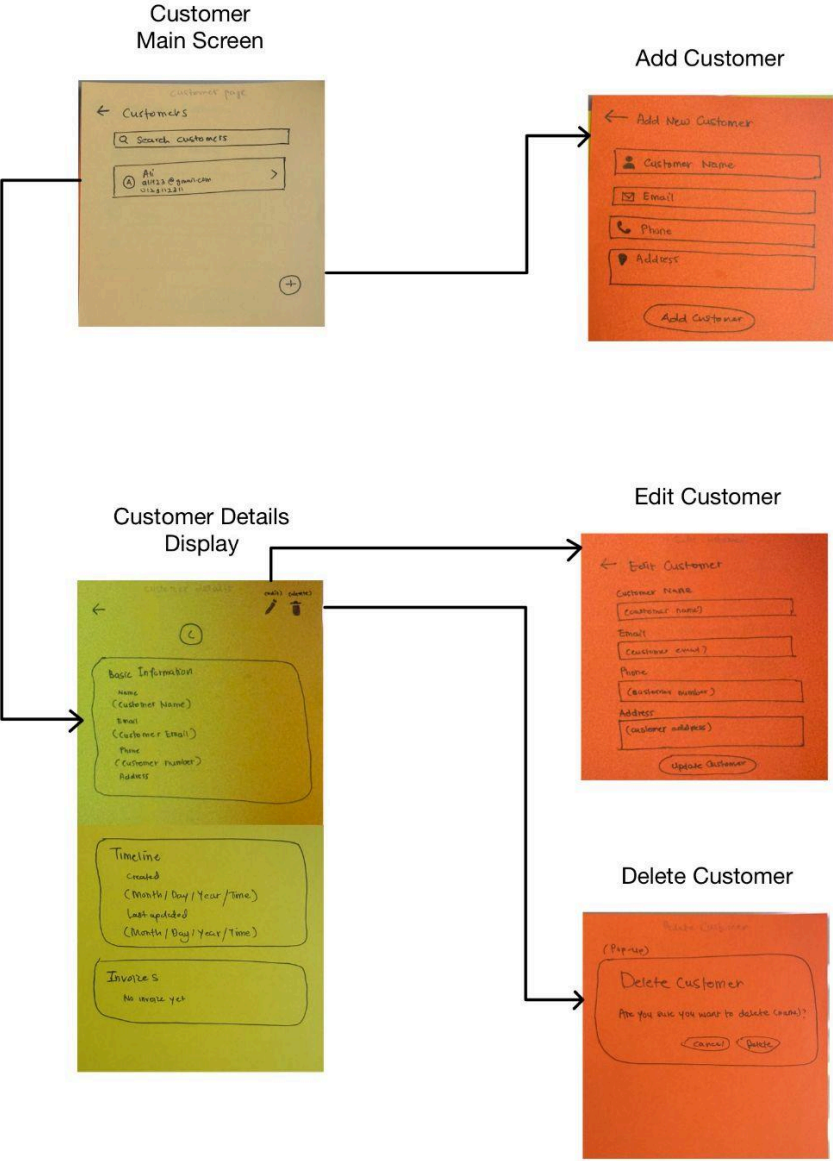
Invoice Management Screen



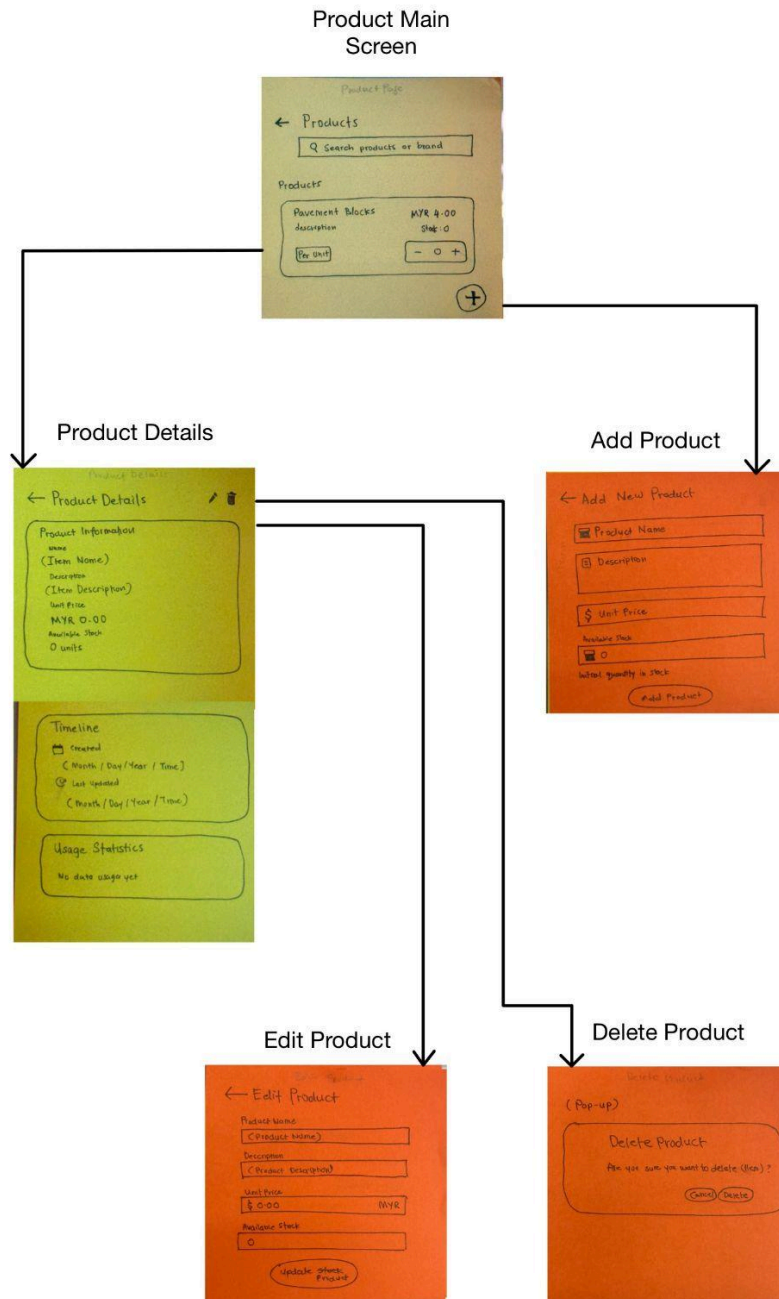
Business Profile Management Screen



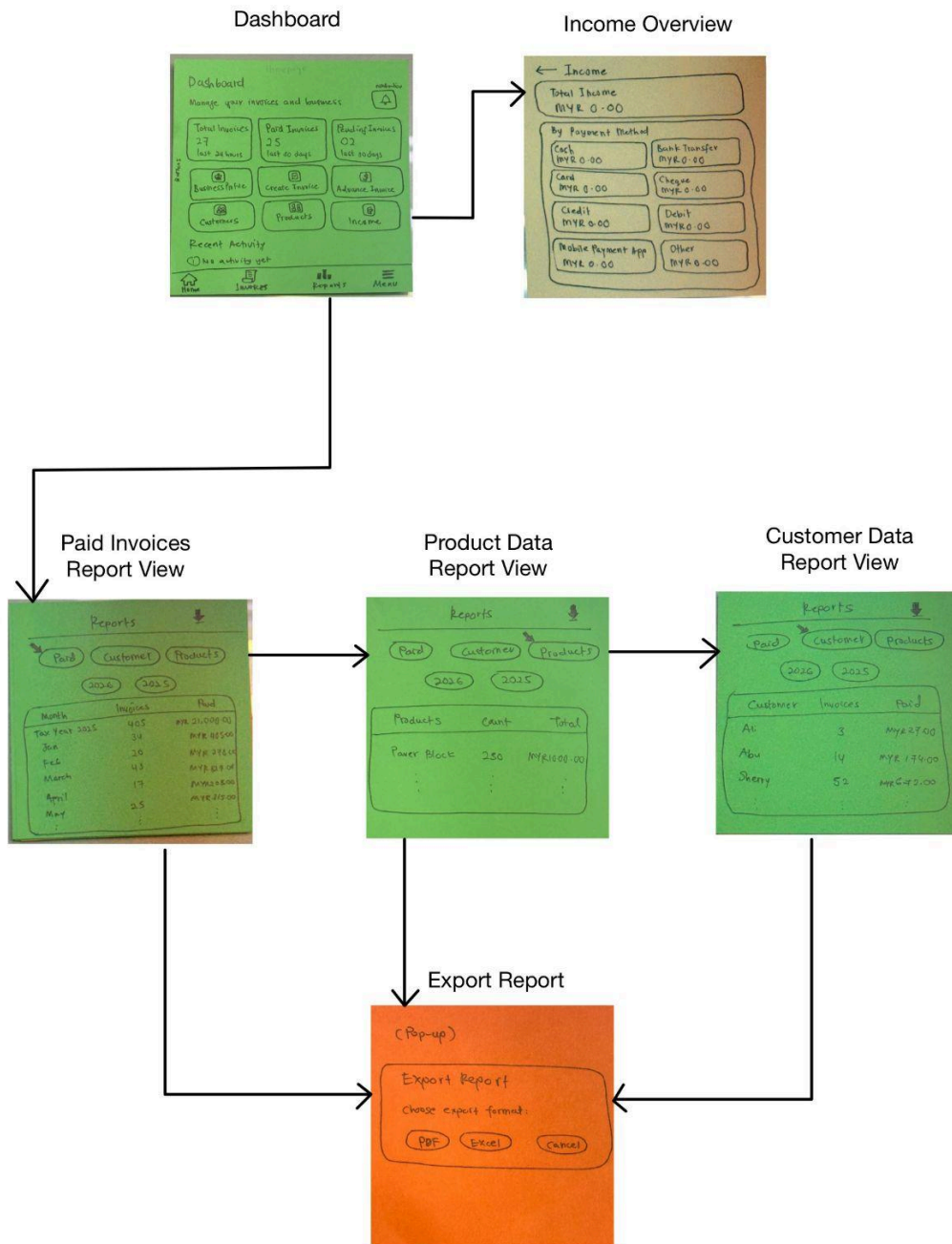
Customer Management Screen



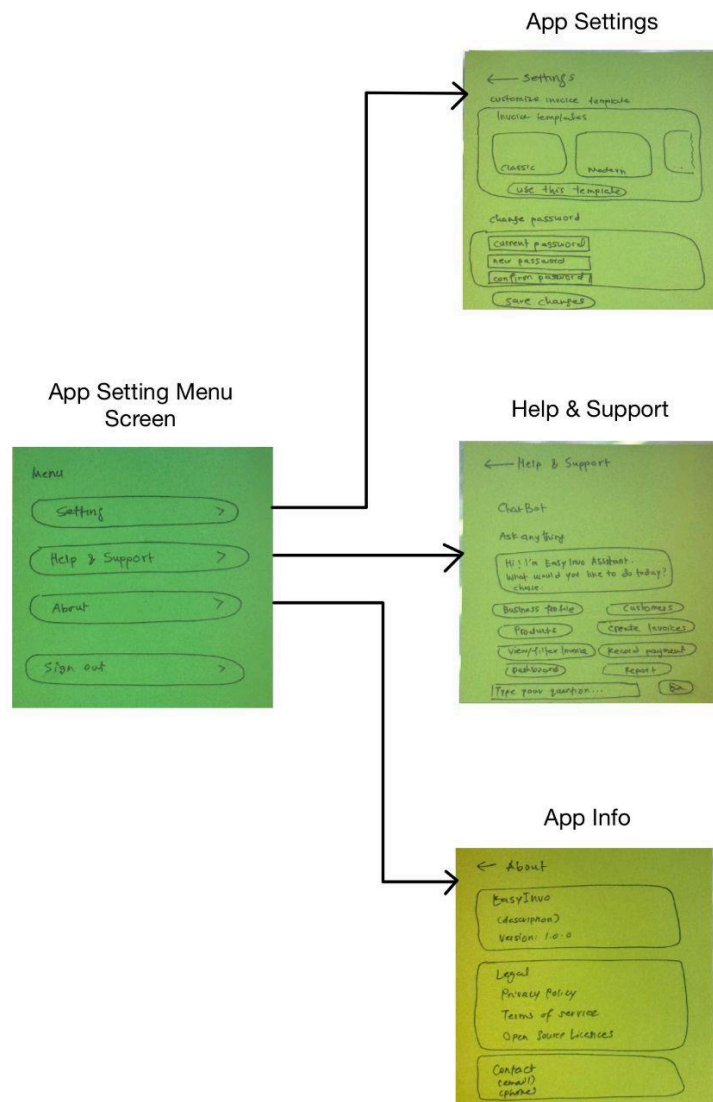
Product Management Screen



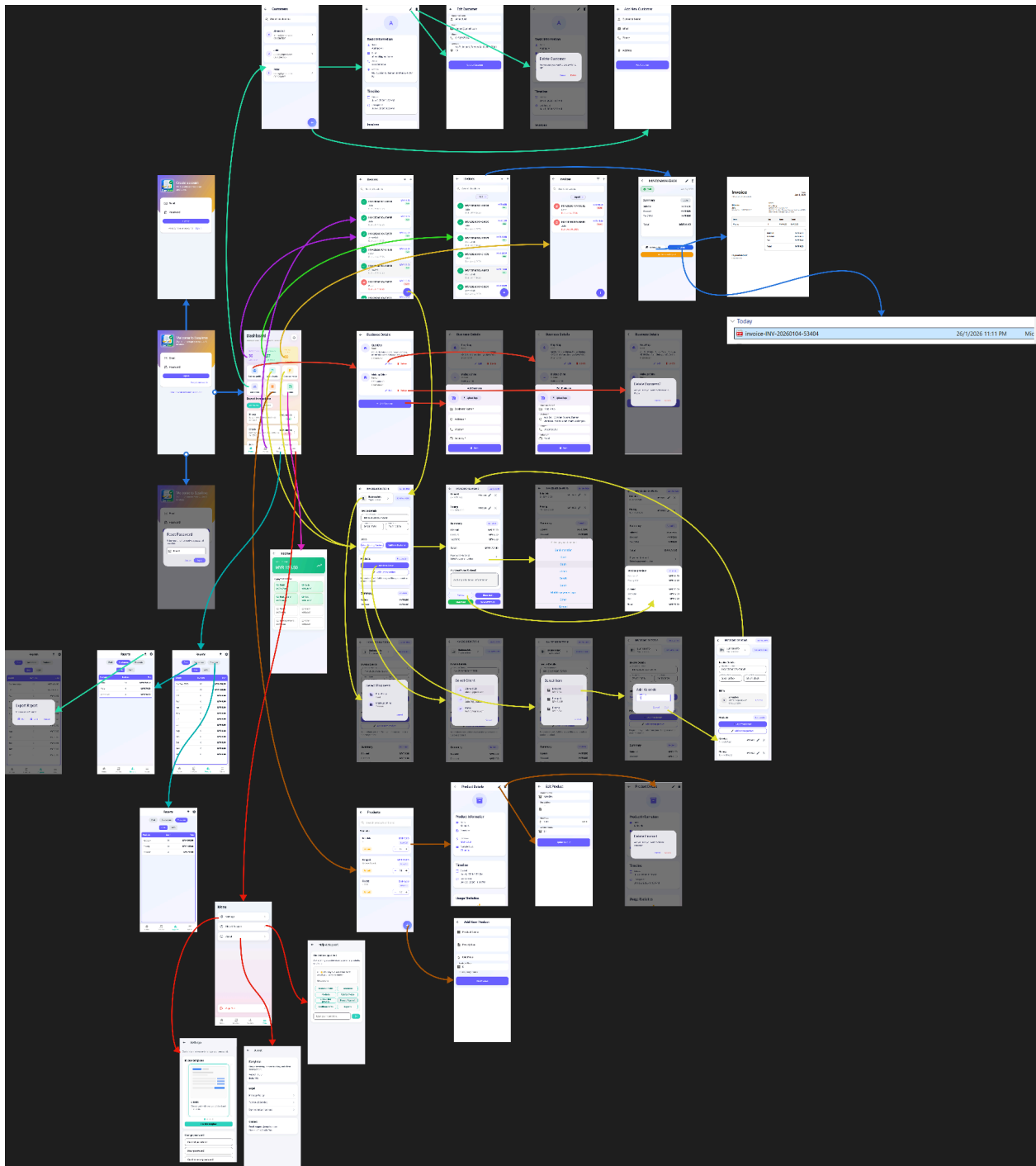
Report Screen



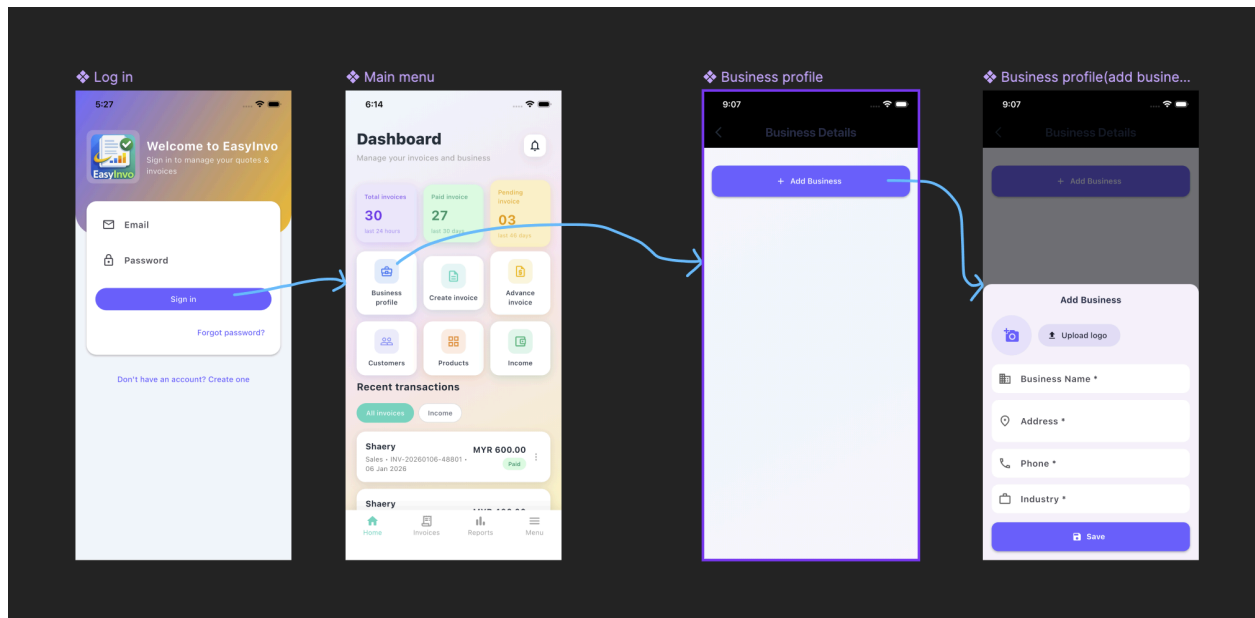
App Setting Screen



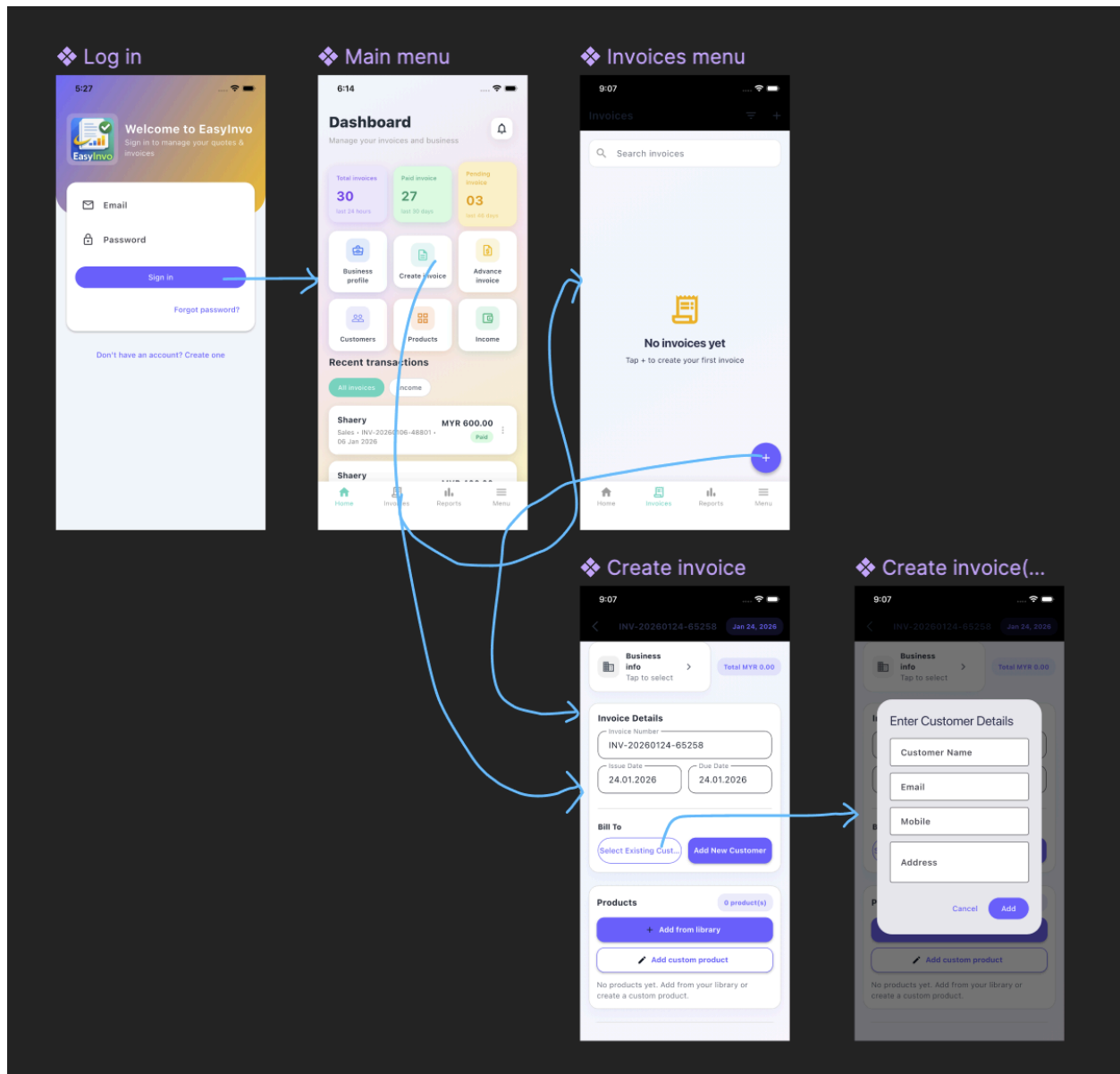
5.2 High Fidelity



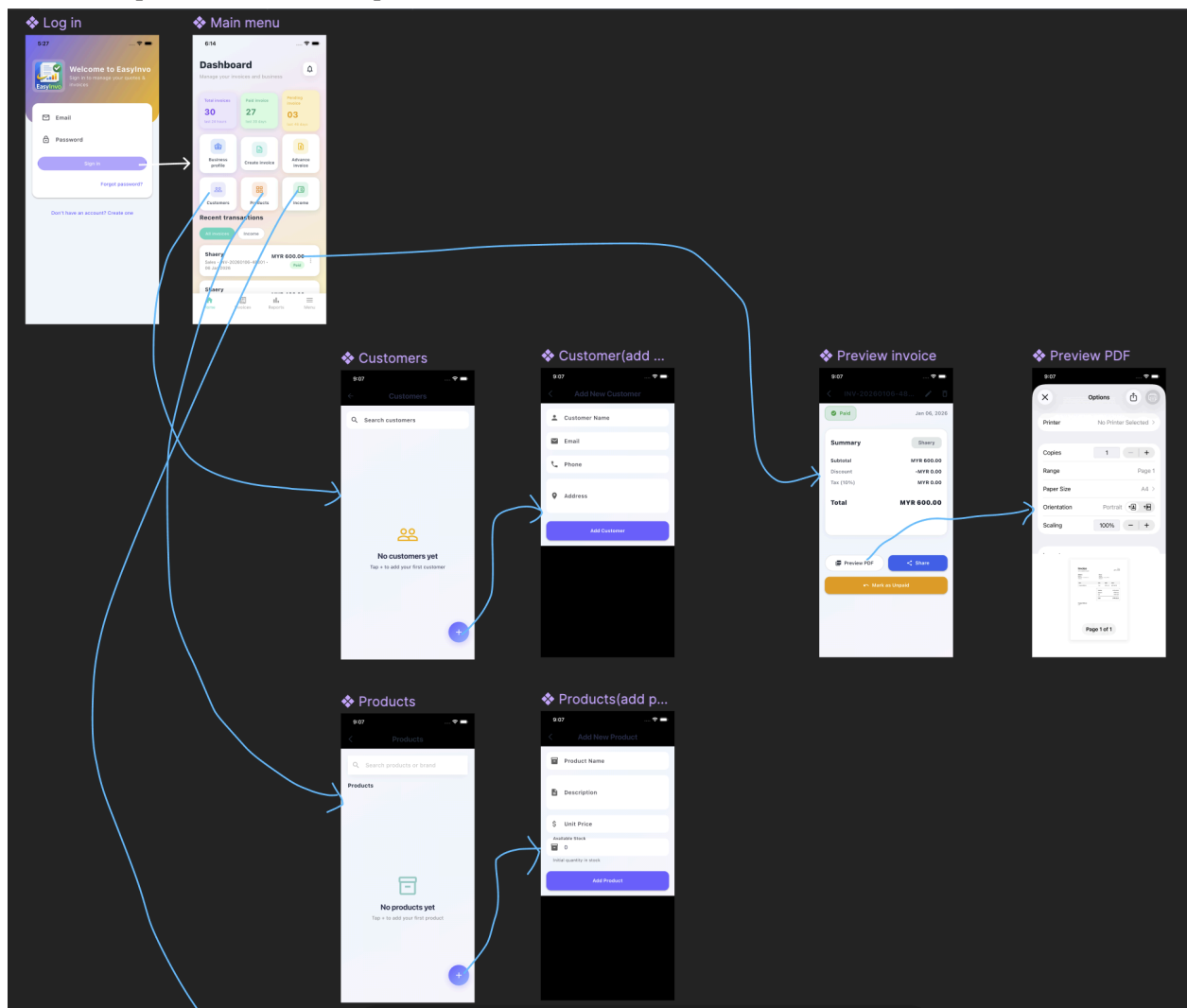
Business profile



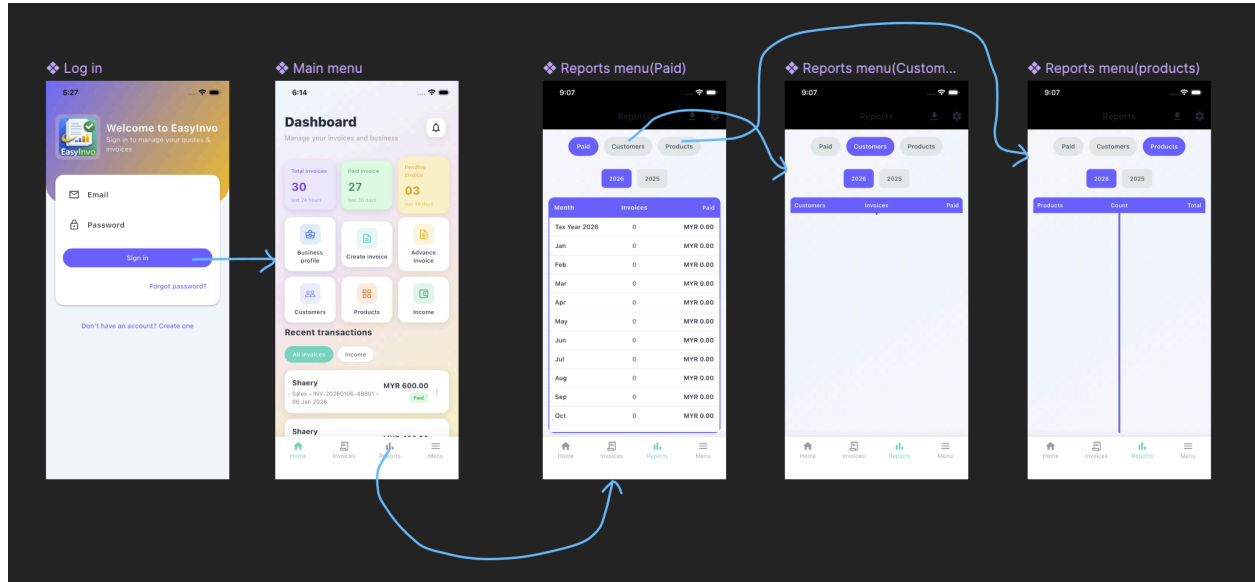
Creating invoice



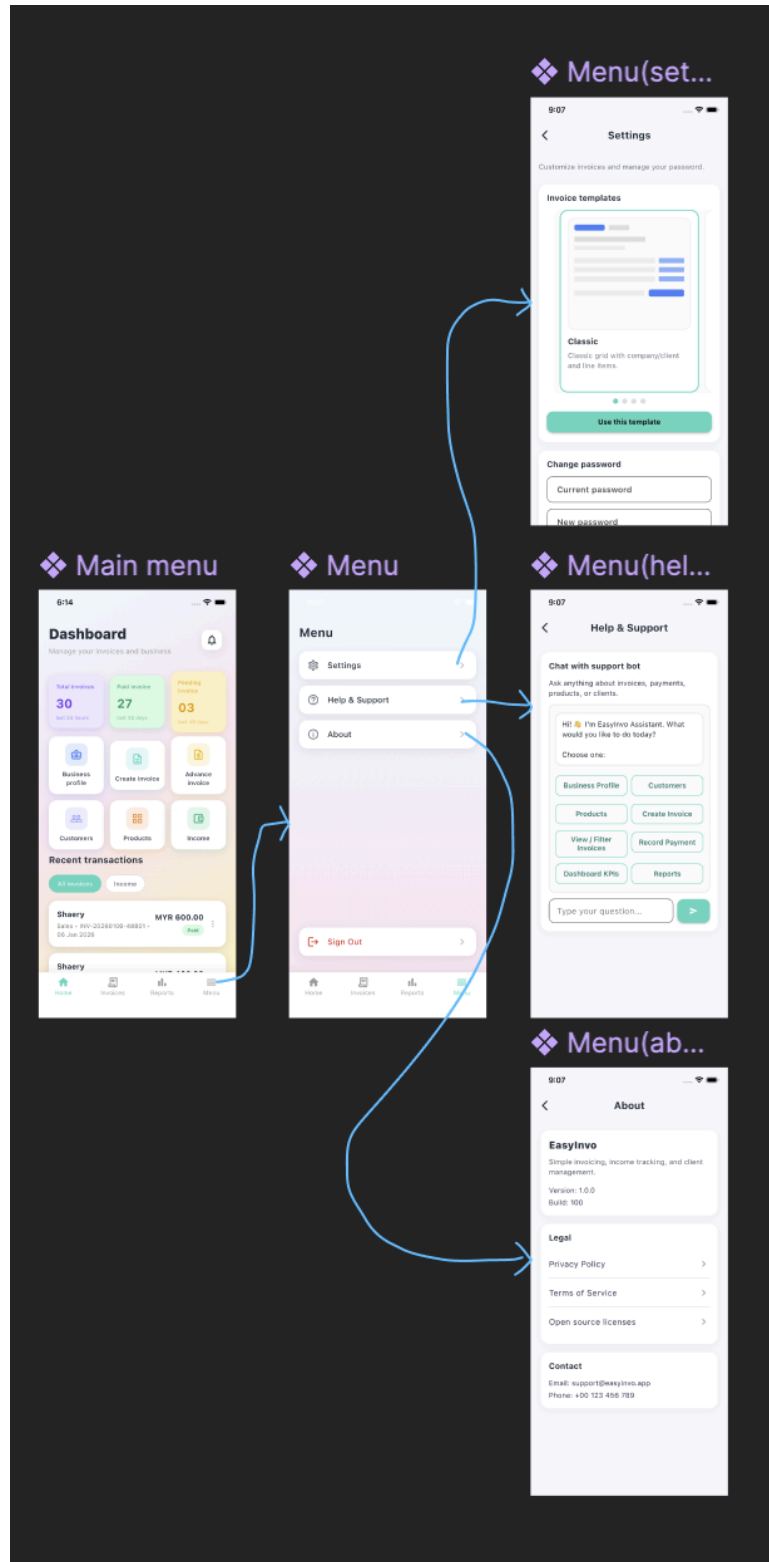
Customers, products and invoice preview



Report menu



Menu settings



6.0 CONCLUSION

How the Design Meets User Needs

- **Simplicity:** The system minimizes steps required to create and send invoices, reducing time and effort.
- **Usability:** Clear navigation and intuitive layouts allow users with minimal technical knowledge to use the system effectively.
- **Accuracy:** Automated calculations for totals and taxes reduce human error.
- **Efficiency:** Dashboard summaries and quick actions improve productivity.
- **Flexibility:** Different user roles ensure appropriate access and security.
- **Professionalism:** High-quality invoice design improves business credibility with clients.

The EasyInvo system design provides a comprehensive, user-centered solution for invoice management. By identifying stakeholders and users, applying UML modeling, designing clear system flows, and developing both low- and high-fidelity interfaces, EasyInvo effectively addresses real user needs. The result is a scalable, efficient, and intuitive invoicing system suitable for small businesses and freelancers.